

Souvik Mandal

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❖ Bengaluru, India

PROFESSIONAL EXPERIENCE

Nanonets

Bengaluru, India

Senior Deep Learning Engineer

Jan. 2024 – Current

- Leading the research and development of the **Nanonets' OCR VLM** series ([Nanonets-OCR-s](#), [Nanonets-OCR2](#)).
 - The open-sourced model on Hugging Face has more than 100k downloads every month. The closed source model is used to process millions of customer documents.
 - Trained the vision language model on more than 10 million **dense documents** to support image markdown, table extraction, and multi-lingual visual question answering.
 - SOTA performance on different benchmarks like OlmOCR, OmniDocBench.
 - Techniques used at different stages: LoRA, full finetuning, GRPO, and model merging.
- Implemented **memory-enabled, on-prem-ready agentic RAG** for 1000+ page documents, powering the Nanonets Agents platform.
- Designed and developed [IDP Leaderboard](#), the most compressive document understanding benchmark.
- Trained and deployed LLMs that **outperform Gemini by 4%** in document information extraction tasks while being **5x more cost-efficient** compared to the previously used models.

Qure.ai

Bengaluru, India

Senior AI Scientist

Mar. 2022 – Dec. 2023

- Developed lung nodule segmentation and characteristics classification models that successfully passed rigorous CE study clearance.
 - Developed **context-based models** that led to a **5% improvement** in segmentation model DICE score.
 - Designed segmentation networks for both CPU and GPU devices, achieving a **runtime of 1 second on CPU and 20 ms on GPU**.
 - Worked on multitask learning to merge characteristics models to reduce processing time.
 - Managed data annotations for lung nodule segmentation and characteristics.
- Worked on medical foundational models
 - Adapted MAE algorithm for self-supervised training on CT scans. Trained a ViT-Large on 60000 nodules in CT scans.
 - Implemented Vision Language models for X-rays. Trained the model on medical reports, X-ray scans and medical ontology datasets.
- Implemented a **training framework** that supports both **2D and 3D volumetric images**. This is used to train the segmentation and classification models.

Fractal Analytics

Bengaluru, India

Data Scientist, AI@SCALE

Sept. 2020 – Mar. 2022

- Designed and implemented IVA-autopilot (semi-supervised, self-supervised, and supervised) which reduces annotated data required for training by 80%, while suggesting which data to annotate in iterations.
- Implemented synthetic dataset generation pipeline for thermal images taken from a drone, which has improved detection model performance by 5%.
- Improved tracking algorithm for drone surveillance in thermal cameras. The MOTA is improved by 1.2% with the implemented algorithm.
- Integrated MLOps tools like triton server, DVC, MLfLow, and Hydra to scale implemented solutions and track project progress.

PUBLICATIONS & PATENTS

- Technical blogs on Deep Learning papers. <https://medium.com/@mandalsouvik>

- Nanonets-OCR2: A model for transforming documents into structured markdown with intelligent content recognition and semantic tagging, Hugging Face, 2025.
- [Validation of a commercially available CAD-system for lung nodule detection and characterization using CT-scans](#). European Radiology, 2024.
- Using an artificial intelligence algorithm to improve radiologists' performance in detecting pulmonary nodules in Chest-CT scans – A multi-reader multi-case study. ECR 2023
- [Needle in a Haystack: Opportunistic Screening of Lung Nodules Amidst Covid-19 Using Deep Learning](#). CHEST 2022
- [Developing an Object and Event Detection and Recognition Module for Autonomous Vehicles Using Simulators](#). ICSCSP, 2020.
- [Linear Approximation and Differential Attacks on Logic Locking Techniques](#). VLSID, 2019.
- [System and methods for efficient machine learning](#). Patent, 2022.

EDUCATION

Indian Institute of Technology, Indore

B. Tech, Computer Science and Engineering

July, 2020

Indore, Madhya Pradesh

- CPI: 8.7/10
- Bachelor thesis advisor: Dr. Gourinath Banda

SKILLS & INTERESTS

Skills: Vision Language Models, Computer Vision, LLMs, Self-supervised learning, medical image analysis.

Interests: Audiobooks, Writing Blogs, Music.